

Set 13: Compounds and formulae

Notes

Chemical symbols and formulae are a shorthand way of communicating information about the structure and composition of elements and compounds. For a molecule the formula indicates what elements and how many of each type are present. So a molecule of carbon dioxide which is represented by the formula CO_2 consists of one carbon atom and two oxygen atoms. For metals, covalent network structures and ionic compounds the formulae indicate the ratio of elements present. So calcium chloride which is represented by the formula CaCl_2 consists of a lattice of positive calcium ions and negative chloride ions. For every calcium ion there are two chloride ions present, but the total number of ions is not known.

Writing the formulae of molecular compounds

Common names are often used for molecular compounds. Some common names and their formulae that you should know are shown in the table (below).

Common Name	Formula
Water	H_2O
Hydrogen peroxide	H_2O_2
Ammonia	NH_3
Methane	CH_4
Ozone	O_3
Hydrogen chloride	HCl
Sulfuric acid	H_2SO_4
Phosphoric acid	H_3PO_4
Nitric acid	HNO_3
Acetic acid	CH_3COOH

For molecular compounds without common names the name of the compound usually includes information about the number of each element present. For example sulfur dioxide is written as SO_2 . Prefixes, listed in the following table, are used to show the number of atoms of each element.

Prefix	Number of atoms
mono-	1
di-	2
tri-	3
tetra-	4
penta-	5
hexa-	6
hepta-	7
octa-	8
nona-	9
deca-	10

Rules for naming molecular compounds:	Example: CO ₂
1. The name of the element closer to the bottom left of the Periodic Table is written first.	Carbon is further to the left than oxygen so is named first
2. The name of the second element is changed to end with the suffix <i>-ide</i>	Oxygen is the second element so will be written as <i>oxide</i>
3. Prefixes are then used to identify the number of atoms of each element in the compound. (If there is only one atom of the first named element no prefix is required.)	Carbon – no prefix needed Oxygen – 2 atoms so use the prefix di - Carbon dioxide

Writing the formulae of ionic compounds

Writing the formula for an ionic compound requires knowledge of the charge on the ions present in the compound. As the overall charge in a compound must be zero, the charges on the ions present must add up to zero. The name only contains the name of the ions. Write the positive ion first.

Examples

1. Writing the formula of sodium fluoride

Ions involved: Na^+ F^-

Use subscript numbers to balance charge $\text{Na}_1^+ \text{F}_1^-$

Charges on the ions add to zero $+1 \quad -1 \quad = 0$



2. Writing the formula of calcium fluoride

Ions involved: Ca^{2+} F^-

Use subscript numbers to balance charge $\text{Ca}_1^{2+} \text{F}_2^-$

Charges on the ions add to zero $+2 \quad -2 \quad = 0$



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Set 13: Exercises

1. Name the following molecular substances
 - (a) HCl
 - (b) SO_2
 - (c) H_2SO_4
 - (d) N_2S
 - (e) P_2Br_4
 - (f) SF_6
2. Write the formula of each of the following molecular compounds.
 - (a) nitrogen monoxide
 - (b) nitrogen dioxide
 - (c) dinitrogen tetroxide
 - (d) sulfur trioxide
 - (e) water
 - (f) pentaphosphorous decoxide
 - (g) hydrogen chloride
 - (h) hydrogen iodide
 - (i) phosphorous tribromide
 - (j) ammonia
3. Write the formula for each of the following ionic compounds.
 - (a) lithium chloride
 - (b) silver iodide
 - (c) potassium nitrate
 - (d) caesium ethanoate
 - (e) barium bromide
 - (f) copper(II) sulfate
 - (g) manganese(IV) oxide
 - (h) nickel nitrate
 - (i) aluminium oxide
 - (j) chromium(III) sulfate
 - (k) lead(IV) phosphate
 - (l) ammonium dichromate
4. Name the following compounds
 - (a) CO
 - (b) N_2O
 - (c) H_2S
 - (d) H_2O_2
 - (e) H_3PO_4
 - (f) Ca_3N_2
 - (g) HNO_3
 - (h) CoHPO_4
 - (i) CuCl
 - (j) FeSO_4